IMPS 2026

Single-Item Reliability for Intensive Longitudinal Data:

A Three-Level Linear Mixed-Effect Autoregressive Modeling Approach

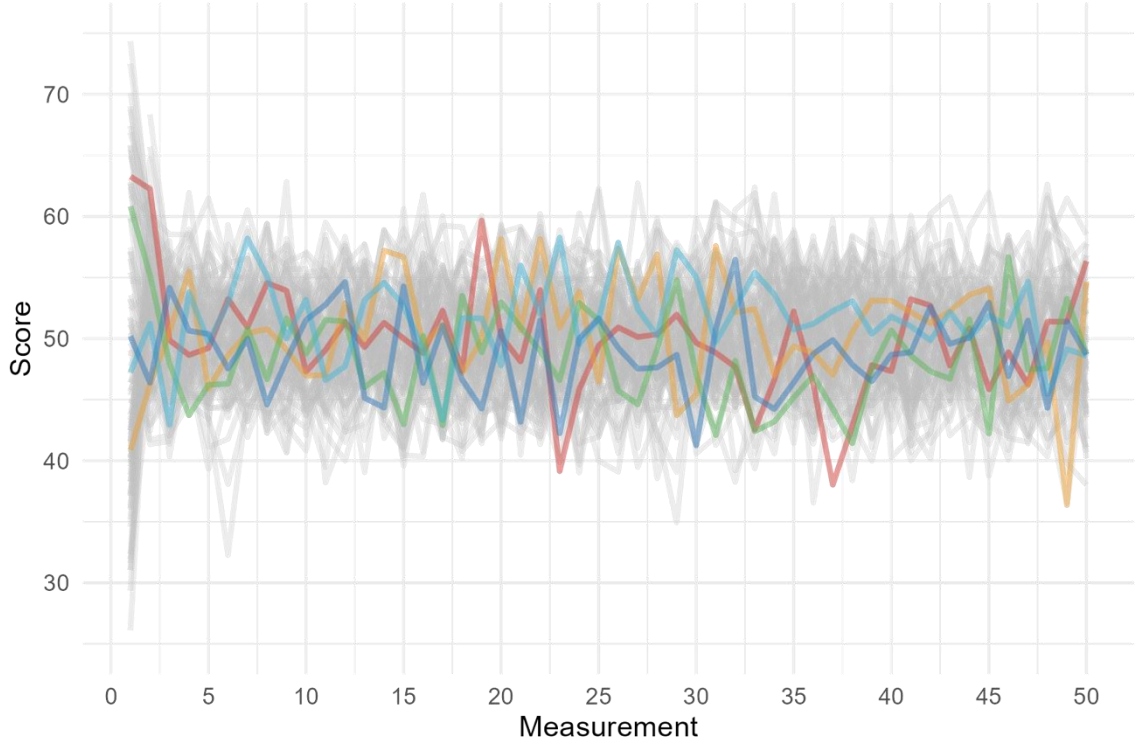
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22/07/2026

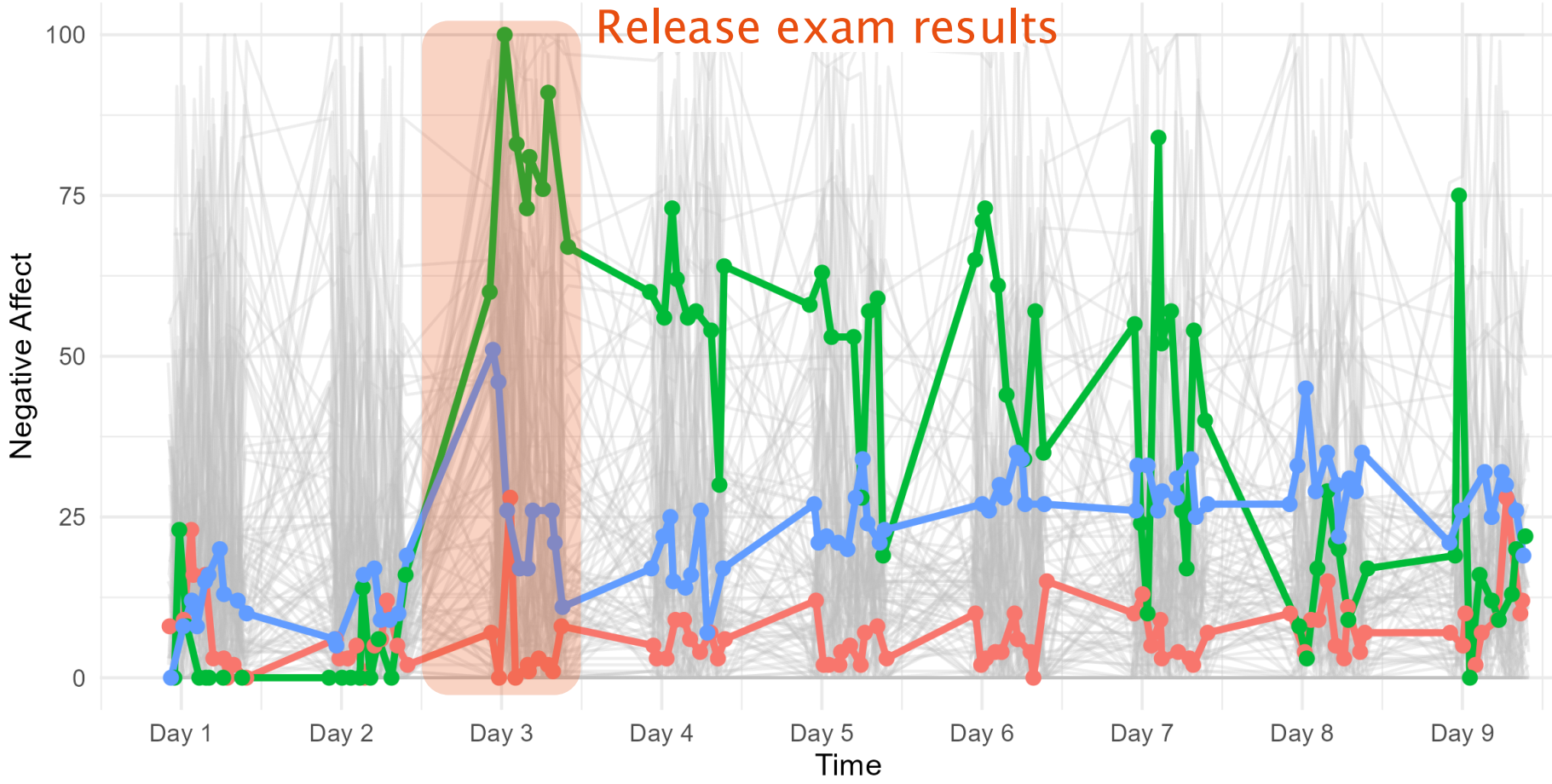
Intensive Longitudinal Data (ILD)

- ILD are a rapidly growing area in psychological study and social sciences focused on understanding processes and behaviors that unfold over time.
- These studies are characterized by
  - high temporal resolution ⇒ modeling intraindividual dynamics
  - collecting multiple individuals ⇒ modeling interindividual differences



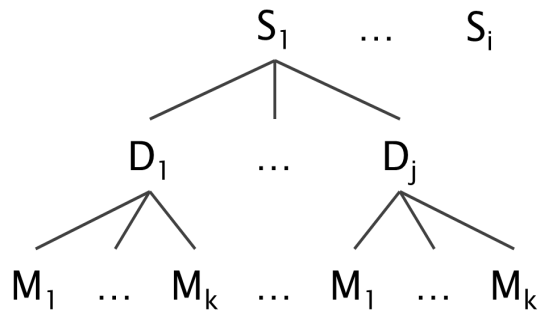
Exam Results Study (Kalokerinos et al., 2019)

- 101 first-year psychology students (87 women; age: $M = 18.64$; $SD = 1.45$)
  - Track for 9 days and collect 10 measurements of their self-report **negative affect**
  - A **single item** scales from 0 (not at all) – 100 (very much)



Properties

1. Observations are often highly correlated
2. Multilevel structure



What is Reliability?

- Classical Test Theory (CTT) (Lord et al., 1968)

$$Y_{ij} = T_{ij} + \varepsilon_{ij} \text{ and } \sigma_Y^2 = \sigma_T^2 + \sigma_\varepsilon^2$$

$$E[Y_{ij}] = T_{ij}$$
$$Cor(\varepsilon_{ij}, \varepsilon_{ij'}) = 0$$
$$Cor(\varepsilon_{ij}, T_{ij}) = 0$$

- The reliability coefficient and **two interpretations**:

$$\rho = \frac{\sigma_T^2}{\sigma_T^2 + \sigma_\epsilon^2}$$

Correlation Approach

(under the parallel assumptions)

$$\rho = Cor(Y_{ij}, Y_{ij'})$$

The degree of similarity between two replicates.

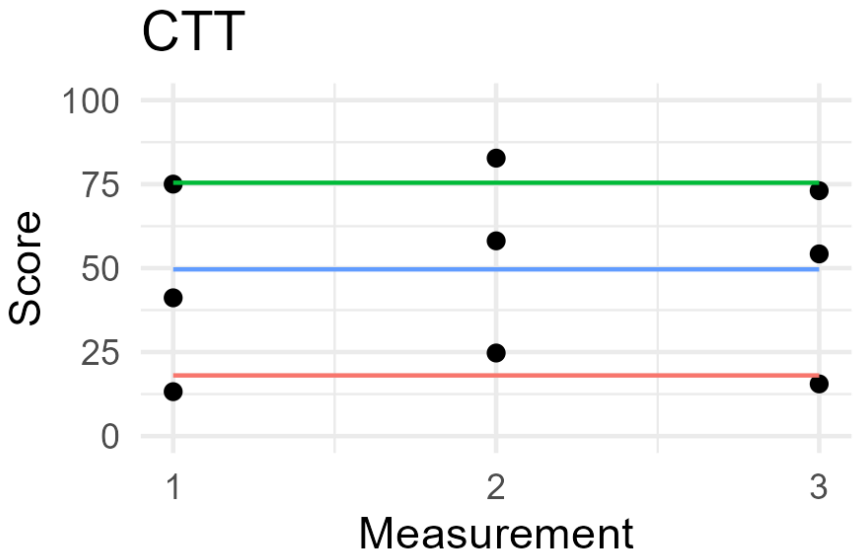
Variance Partitioning Approach

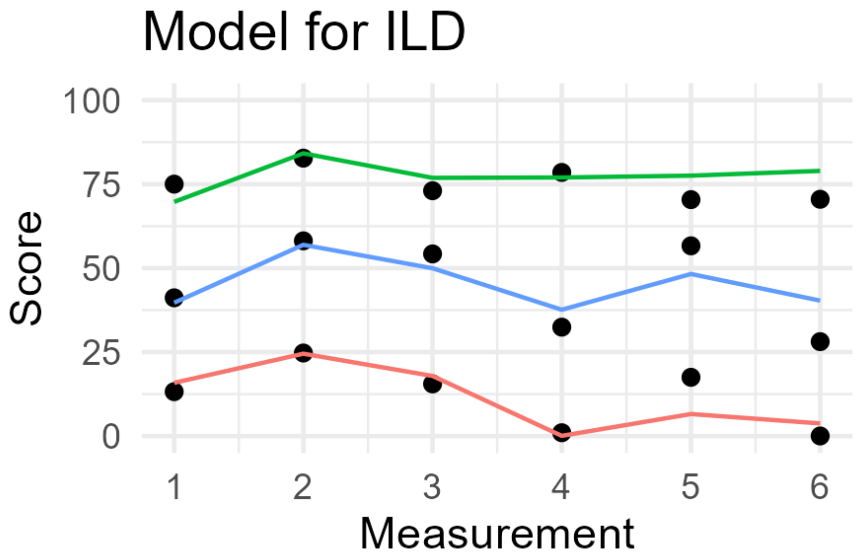
$$\rho = 1 - \frac{\sigma^2_{\text{unsys.}}}{\sigma^2_{\text{total}}} = \frac{\sigma^2_{\text{sys.}}}{\sigma^2_{\text{sys.}} + \sigma^2_{\text{unsys.}}}$$

How well can we distinguish targets from error?

Limitations of CTT

1. ILD have a more complicated structure
   - Person's true score changes over time
   - Observations are usually **correlated** (the parallel assumptions do not hold)
   - **Nested structure**: between-person variance vs. within-person variance

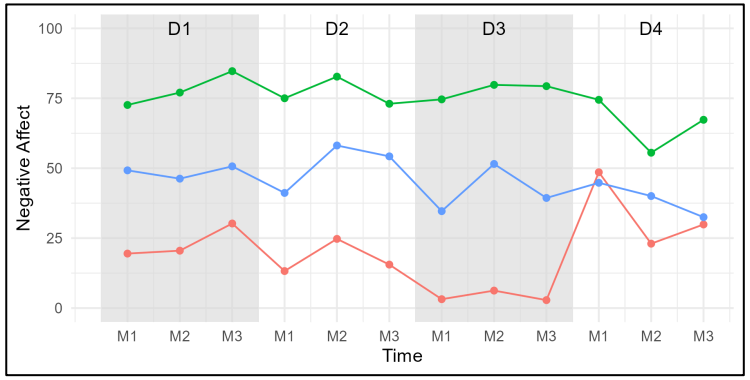


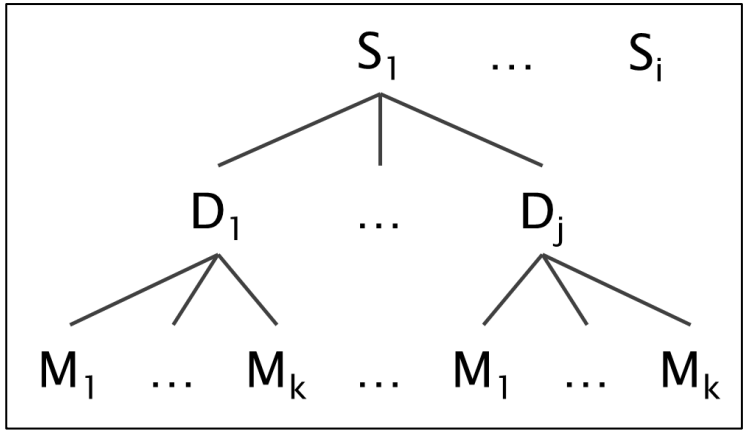


2. Multiple identifiable sources of factors/errors are not accounted for
   - E.g., gender, item, day effects, etc.

Goals of This Project

Data





Model

Account for

the **serial correlation**

and

the **multilevel structure**
(S > D > M)

Reliability

Correlation Approach

$$\rho = Cor(Y_{ij}, Y_{ij'})$$

Variance Partitioning Approach

$$R = 1 - \frac{\sigma^2_{\text{unsys.}}}{\sigma^2_{\text{total}}}$$

Three-Level LMM with Two AR(1) Processes

L3: Sub-level L2: Day-level L1: Moment-level

Subject: $i = 1, \dots, N$
Day: $j = 1, \dots, D$
Moment: $k = 1, \dots, M$

$$y_{ijk} = \beta + s_i + d_{ij} + \nu_{ij} + \omega_{ijk} + \epsilon_{ijk}$$

Grand mean

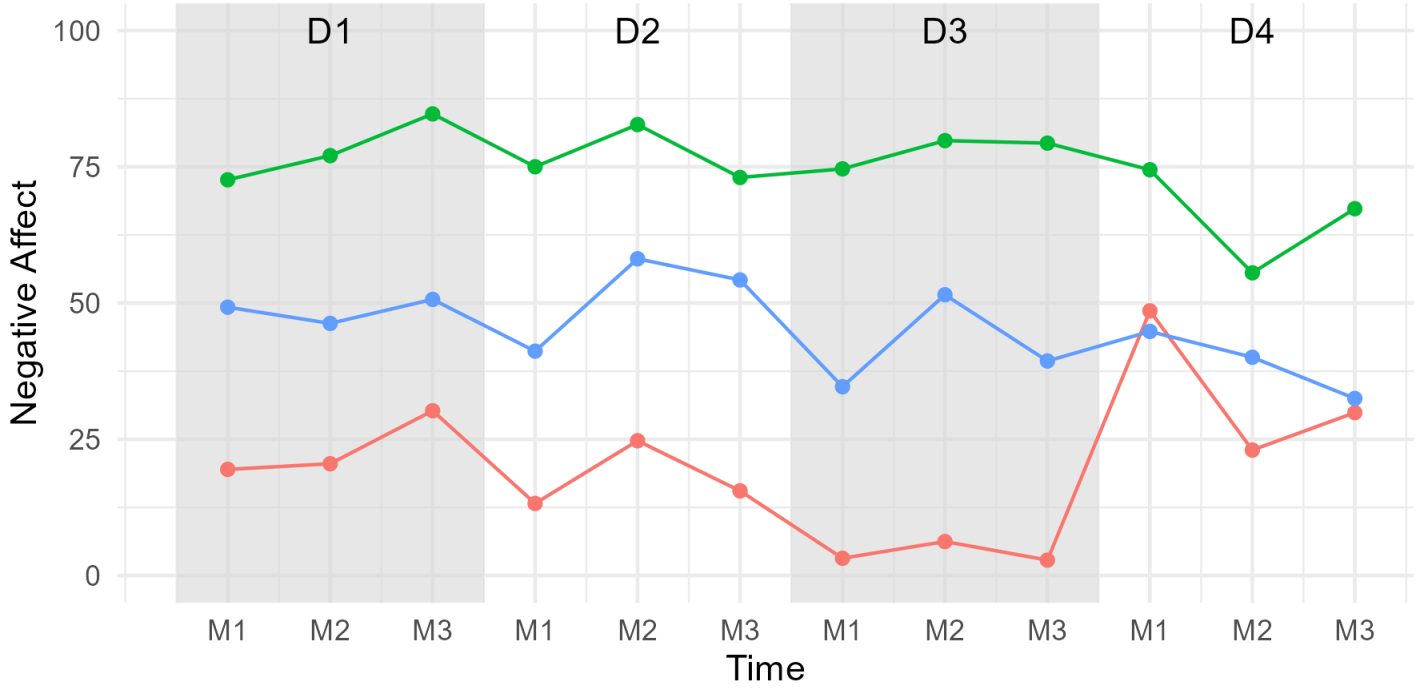
Subject effect
$s_i \sim N(0, \sigma_s^2)$

Day effect
$d_{ij} \sim N(0, \sigma_d^2)$

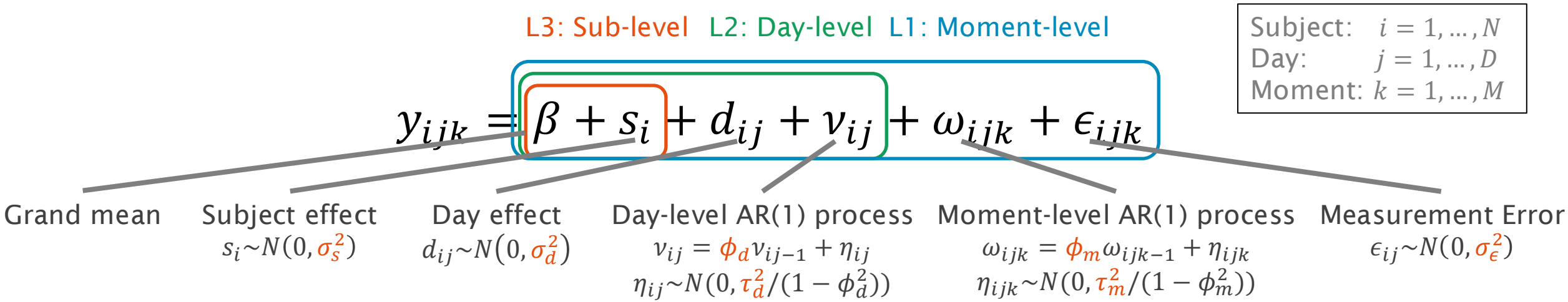
Day-level AR(1) process
$\nu_{ij} = \phi_d \nu_{ij-1} + \eta_{ij}$
$\eta_{ij} \sim N(0, \tau_d^2/(1-\phi_d^2))$

Moment-level AR(1) process
$\omega_{ijk} = \phi_m \omega_{ijk-1} + \eta_{ijk}$
$\eta_{ijk} \sim N(0, \tau_m^2/(1-\phi_m^2))$

Measurement Error
$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$



Three-Level LMM with two AR(1) Processes



Total Variance

$$\mathbf{V} = \mathrm{Var}(\boldsymbol{y_i}) = \boldsymbol{\Sigma_S} + \boldsymbol{\Sigma_D} + \boldsymbol{\Sigma_\nu} + \boldsymbol{\Sigma_\omega} + \boldsymbol{\Sigma_\epsilon}$$

Person-Specific Variance

$$\mathbf{V_i} = \mathrm{Var}(\boldsymbol{y_i} \mid i) = \boldsymbol{\Sigma_D} + \boldsymbol{\Sigma_\nu} + \boldsymbol{\Sigma_\omega} + \boldsymbol{\Sigma_\epsilon}$$

Two Approaches to Defining Reliability

Correlation Approach

$$\rho = Cor(Y_{ij}, Y_{ij'})$$

Variance Partitioning Approach

$$R = 1 - \frac{\sigma^2_{\text{unsys.}}}{\sigma^2_{\text{total}}} = \frac{\sigma^2_{\text{sys.}}}{\sigma^2_{\text{sys.}} + \sigma^2_{\text{unsys.}}}$$

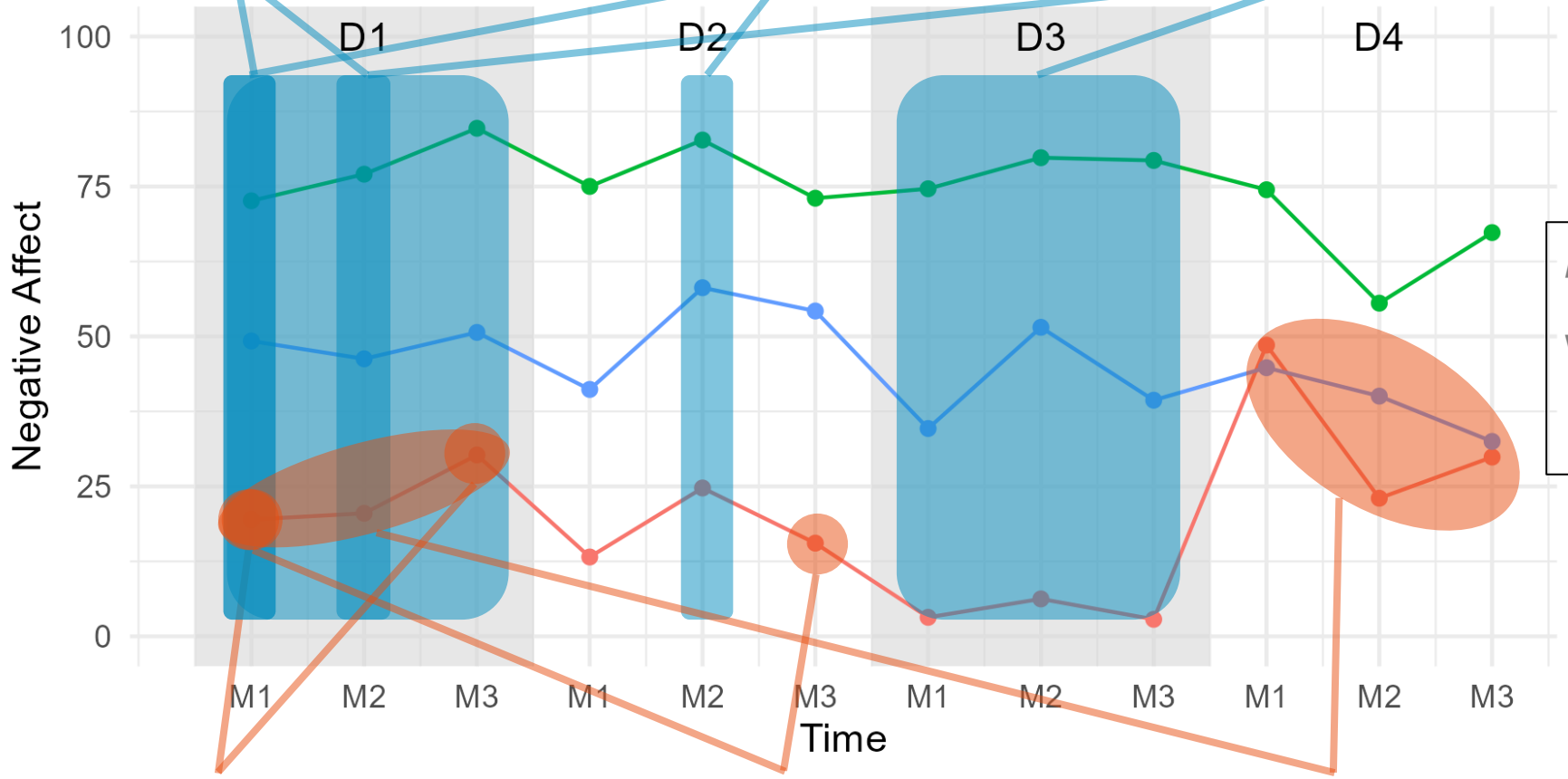
- Vangeneugden et al. (2004, 2005)
  - Define reliability for an LMM with one serial correlation
- Lin et al. (2025)
  - Separate reliability at different hierarchical levels

$$y_{ijk} = \beta + s_i + d_{ij} + \nu_{ij} + \omega_{ijk} + \epsilon_{ijk}$$

$$Cor(y_{ijk}, y_{ij'k'}) = ?$$

Reliability: Correlation Approach

$\rho^B_M = Cor(y_{ijk}, y_{ijk'})$ $\rho^B_D = Cor(y_{ijk}, y_{ij'k'})$ $\rho^B_{aveD} = Cor(\bar{y}_{ij\cdot}, \bar{y}_{ij'\cdot})$



ρ: reliability
B: between-person
W: within-person
M: mom.-to-mom.
D: day-to-day

$\rho^W_M = Cor(y_{ijk}, y_{ijk'} | i)$ $\rho^W_D = Cor(y_{ijk}, y_{ij'k'} | i)$ $\rho^W_{aveD} = Cor(\bar{y}_{ij\cdot}, \bar{y}_{ij'\cdot} | i)$

Reliability: Correlation Approach

Moment-to-Moment

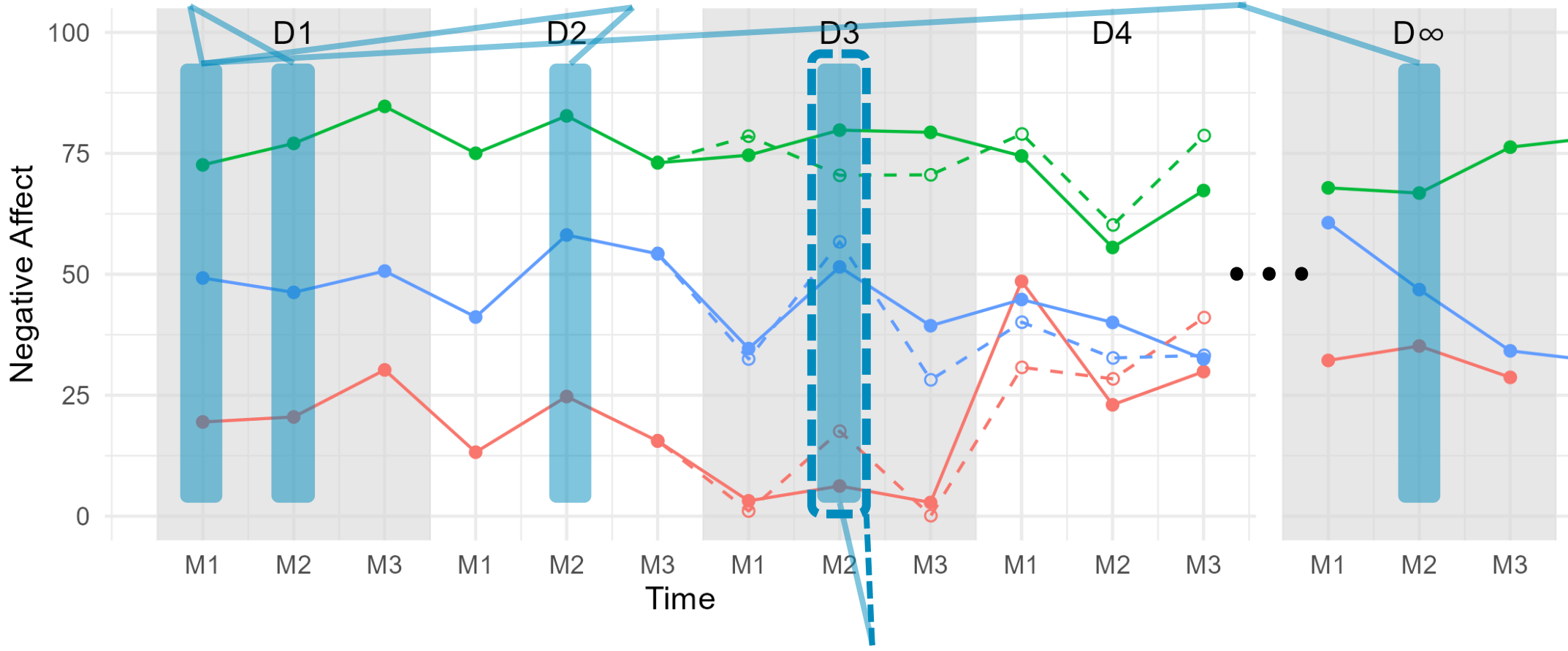
$\rho_M^B = Cor(y_{ijk}, y_{ijk'})$

Day-to-Day

$\rho_D^B = Cor(y_{ijk}, y_{ij'k'})$

Stability

$\rho_\infty^B = Cor(y_{ijk}, y_{i\infty\infty})$



Instantaneity

$\rho_0^B = Cor(y_{ijk}, y'_{ijk})$

Relationship Between Two Approach

Between-person	Time span			
	Instantaneity	Moment-to-Moment	Day-to-Day	Stability
M1: Three-level LMM with two AR(1)	$\rho_0^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_M^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 \phi_m^{\lvert k-k' \rvert}}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_D^B = \frac{\sigma_s^2 + \tau_d^2 \phi_d^{\lvert j-j' \rvert}}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_\infty^B = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$

Within-person	Time span			
	Instantaneity	Moment-to-Moment	Day-to-Day	Stability
M1: Three-level LMM with two AR(1)	$\rho_0^W = \frac{\sigma_d^2 + \tau_d^2 + \tau_m^2}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_M^W = \frac{\sigma_d^2 + \tau_d^2 + \tau_m^2 \phi_m^{\lvert k-k' \rvert}}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_D^W = \frac{\tau_d^2 \phi_d^{\lvert j-j' \rvert}}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	-

Two Approaches to Defining Reliability

Correlation Approach

$$\rho = Cor(Y_{ij}, Y_{ij'})$$

- Vangeneugden et al. (2004, 2005)
  - Define reliability for an LMM with one serial correlation
- Lin et al. (2025)
  - Separate reliability at different hierarchical levels

$$y_{ijk} = \beta + s_i + d_{ij} + \nu_{ij} + \omega_{ijk} + \epsilon_{ijk}$$

$$Cor(Y_{ijk}, Y_{ij'k'}) = ?$$

Variance Partitioning Approach

$$R = 1 - \frac{\sigma^2_{\text{unsys.}}}{\sigma^2_{\text{total}}} = \frac{\sigma^2_{\text{sys.}}}{\sigma^2_{\text{sys.}} + \sigma^2_{\text{unsys.}}}$$

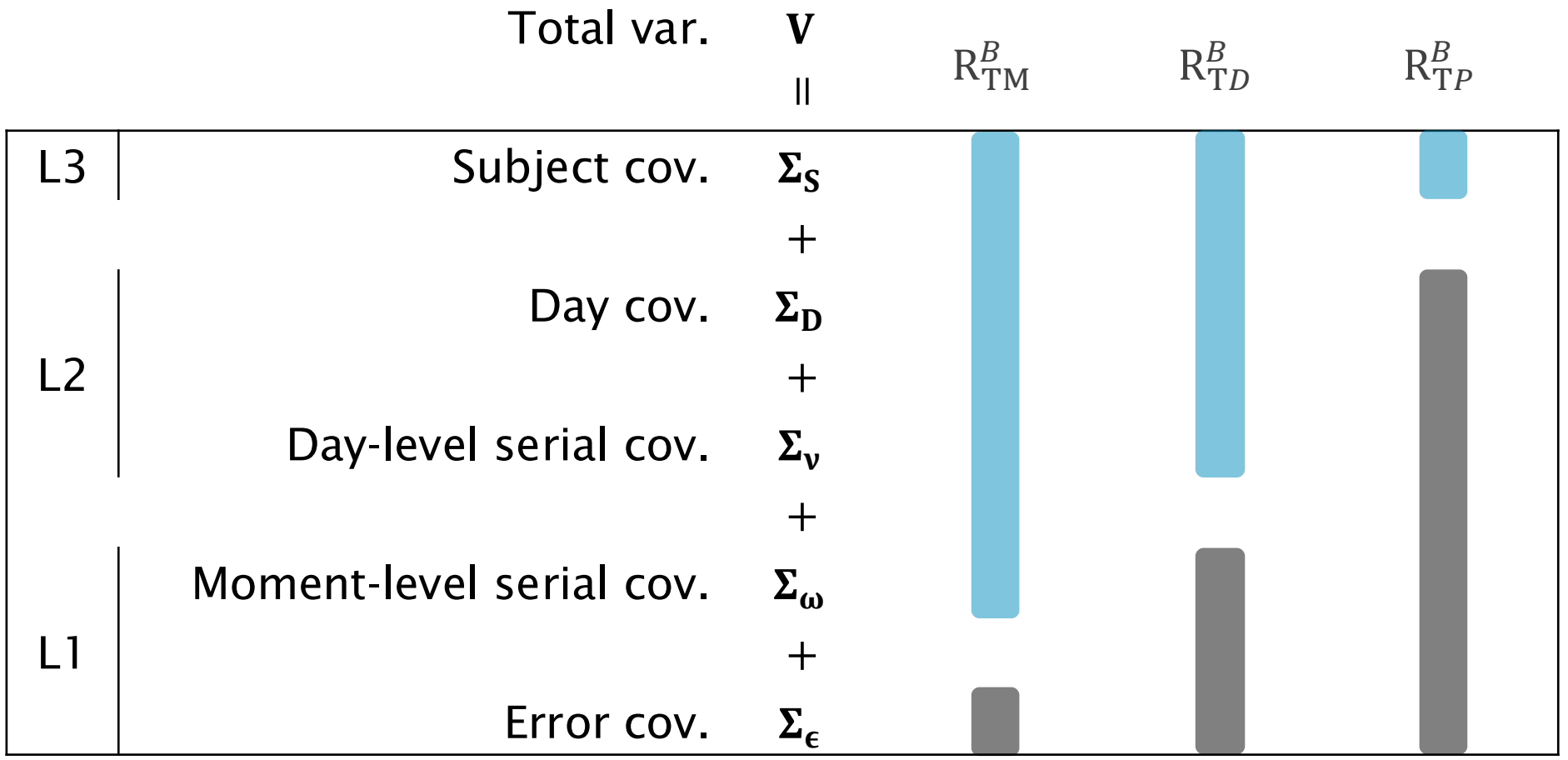
- Laenen et al, (2007, 2009)
  - Define reliability for an LMM with one serial correlation in the vectorized representation

$$R_T = 1 - \frac{tr(\boldsymbol{\Sigma}_{\text{unsys}})}{tr(\mathbf{V})}$$

$$\mathbf{V} = \boldsymbol{\Sigma}_{\mathbf{S}} + \boldsymbol{\Sigma}_{\mathbf{D}} + \boldsymbol{\Sigma}_{\boldsymbol{\nu}} + \boldsymbol{\Sigma}_{\boldsymbol{\omega}} + \boldsymbol{\Sigma}_{\boldsymbol{\epsilon}}$$

What is the unsystematic part $\boldsymbol{\Sigma}_{\text{unsys}}$=?

Reliability: Variance Partitioning Approach



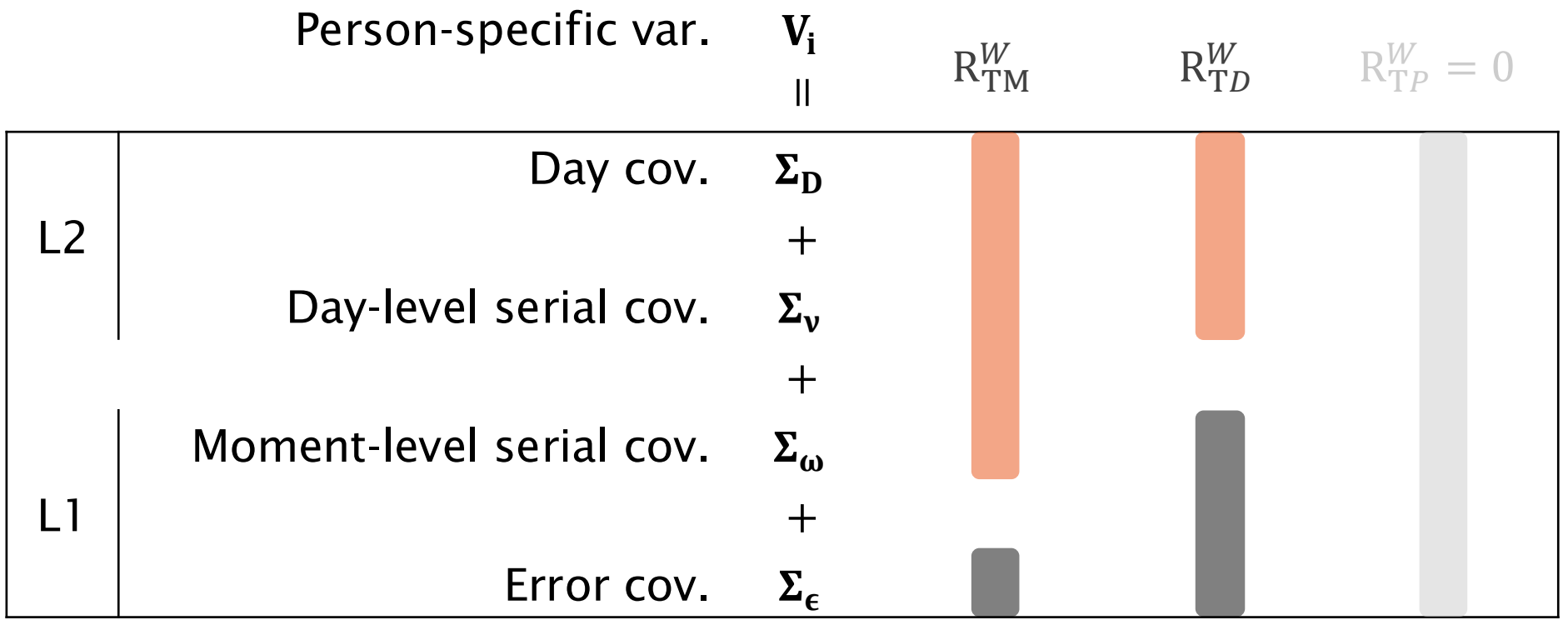
$$R_T = 1 - \frac{tr(\boldsymbol{\Sigma}_{\mathbf{unsys}})}{tr(\mathbf{V})}$$

Systematic part
$\mathbf{V} - \boldsymbol{\Sigma}_{\mathbf{unsys}}$

Unsystematic part
$\boldsymbol{\Sigma}_{\mathbf{unsys}}$

R: reliability
B: between-person
W: within-person
M: moment
D: day
P: subject

Reliability: Variance Partitioning Approach



$$R_{T\,i} = 1 - \frac{tr(\mathbf{\Sigma_{unsys_i}})}{tr(\mathbf{V_i})}$$

Systematic part
$V_i - \mathbf{\Sigma_{unsys_i}}$

Unsystematic part
$\mathbf{\Sigma_{unsys_i}}$

R: reliability
B: between-person
W: within-person
M: moment
D: day
P: subject

Relationship Between Two Approaches

Between-person	Time span			
	Instantaneity	**Moment-to-Moment**	**Day-to-Day**	**Stability**
M1: Three-level LMM with two AR(1)	$\rho_0^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2} = \mathrm{R}_{TM}^B$	$\rho_M^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 \phi_m^{\lvert k-k' \rvert}}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_D^B = \frac{\sigma_s^2 + \tau_d^2 \phi_d^{\lvert j-j' \rvert}}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_\infty^B = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2} = \mathrm{R}_{TP}^B$

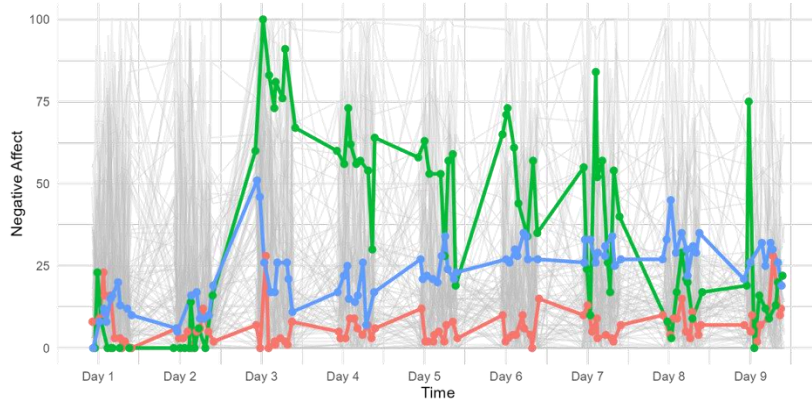
$$\mathrm{R}_{TD}^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$$

Within-person	Time span			
	Instantaneity	**Moment-to-Moment**	**Day-to-Day**	**Stability**
M1: Three-level LMM with two AR(1)	$\rho_0^W = \frac{\sigma_d^2 + \tau_d^2 + \tau_m^2}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2} = \mathrm{R}_{TM}^W$	$\rho_M^W = \frac{\sigma_d^2 + \tau_d^2 + \tau_m^2 \phi_m^{\lvert k-k' \rvert}}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_D^W = \frac{\tau_d^2 \phi_d^{\lvert j-j' \rvert}}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	-

$$\mathrm{R}_{TD}^W = \frac{\sigma_d^2 + \tau_d^2}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$$

Illustrative Example

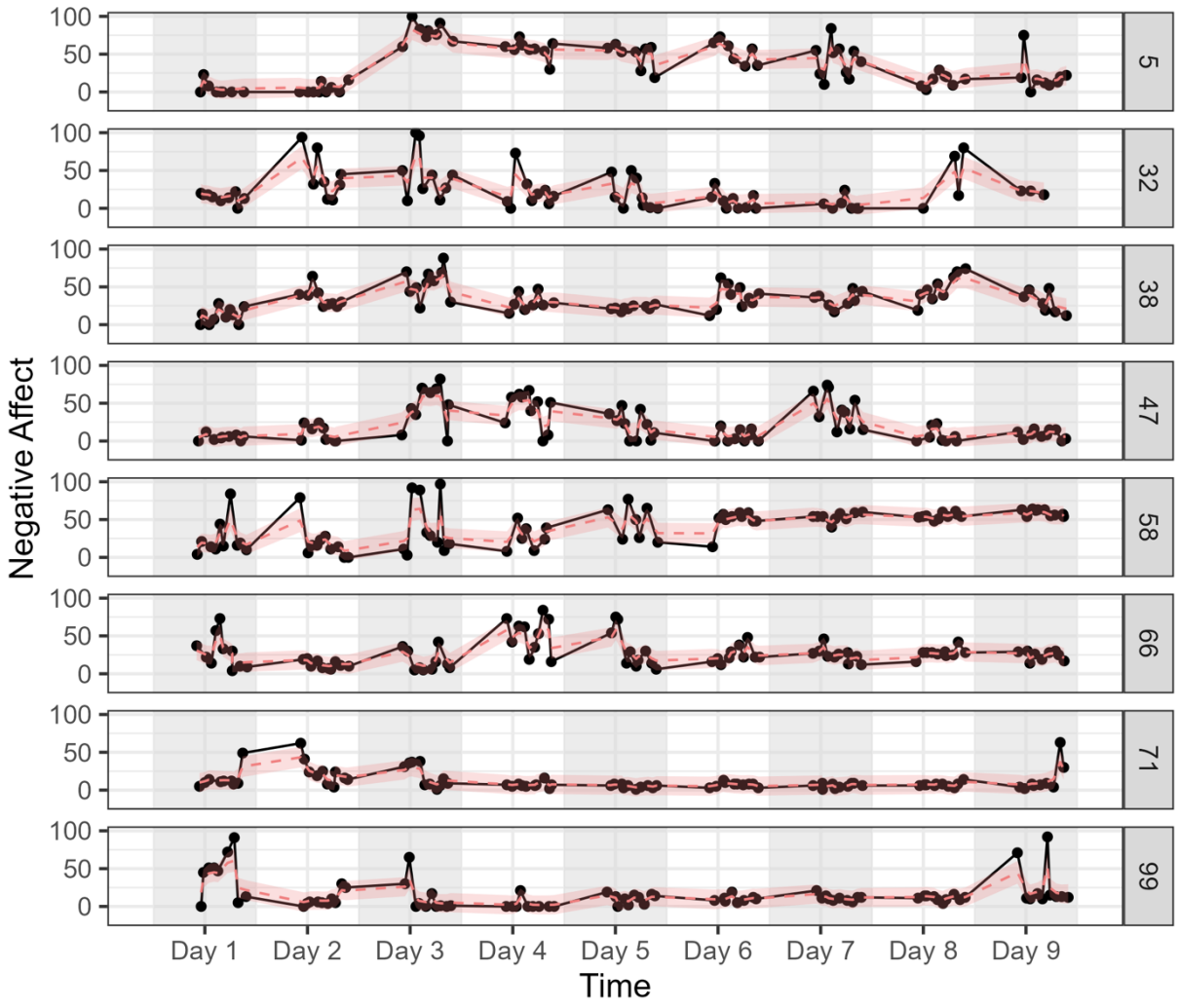
- Data
  - Exam Result Study (Kalokerinos et al., 2019)
  - Negative affect: self-report, single item, 0-100
  - Compliance rate: mean=90.8%, SD = 7.4%, min = 64.4%



- Methods
  - Bayesian inference
  - Run 4 parallel MCMC chains by using Stan
    - In each chain: #burn-in = 32000, #post-warm-up = 32000, #thin = 8
    - Total draws of #post-warm-up = 16000
    - All Rhat of parameters ≤ 1.015
    - The minimum effective sample size (ESS) is 269.5
  - Missing values are seen as the unknown parameters

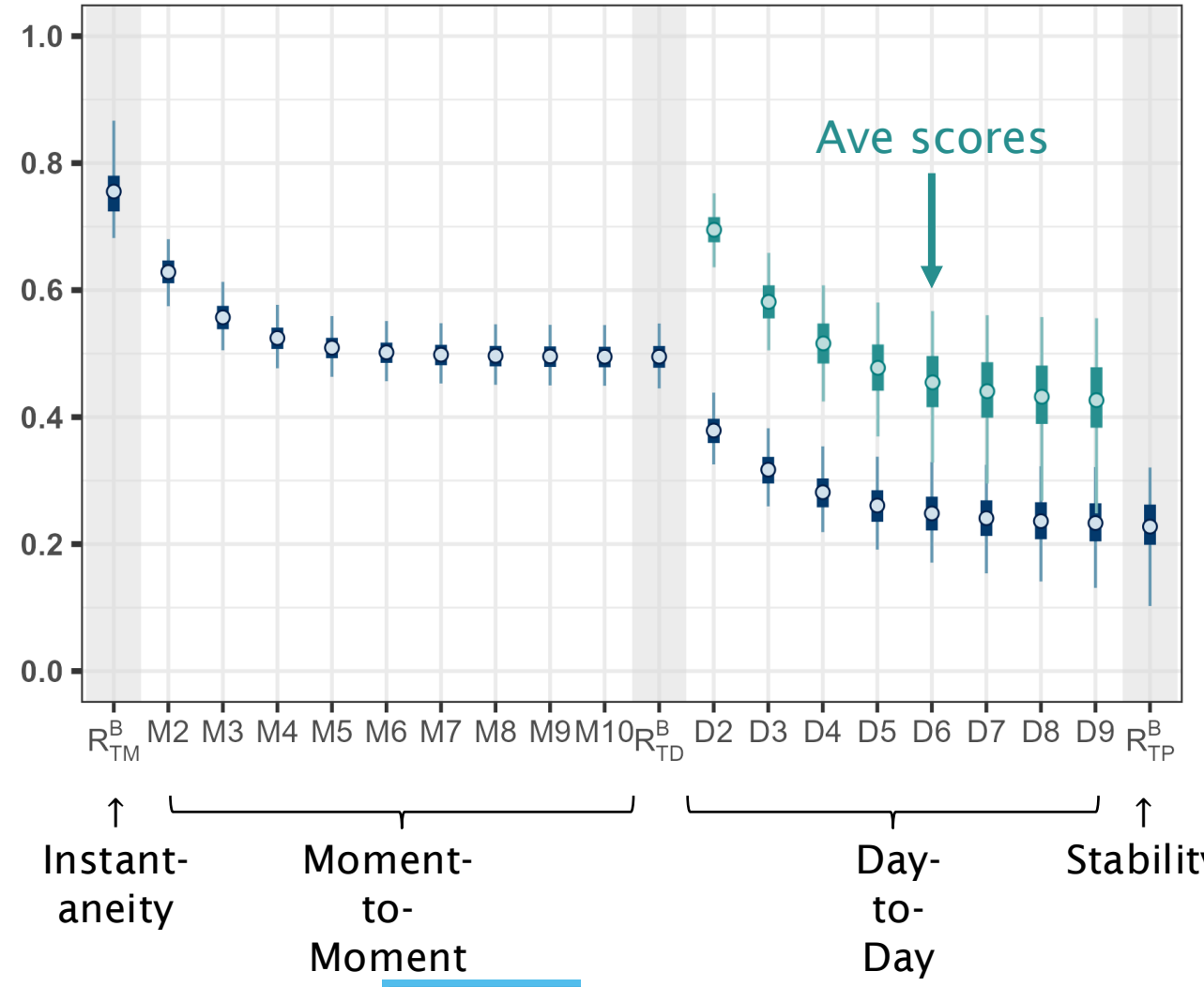
Fitting results

Parameter	Posterior Mean (SD)	95% HDI
Fixed effect		
β	19.48 (1.26)	[17.04, 21.57]
Autoregressive effect		
ϕ_d	0.57 (0.08)	[0.46, 0.71]
ϕ_m	0.45 (0.08)	[0.32, 0.58]
Standard deviation		
σ_s	10.39 (1.60)	[7.95, 12.56]
σ_d	1.49 (1.08)	[0.13, 3.55]
τ_d	11.17 (0.88)	[9.98, 12.69]
τ_m	11.16 (0.84)	[9.96, 12.71]
σ_ϵ	10.80 (1.04)	[8.82, 12.10]

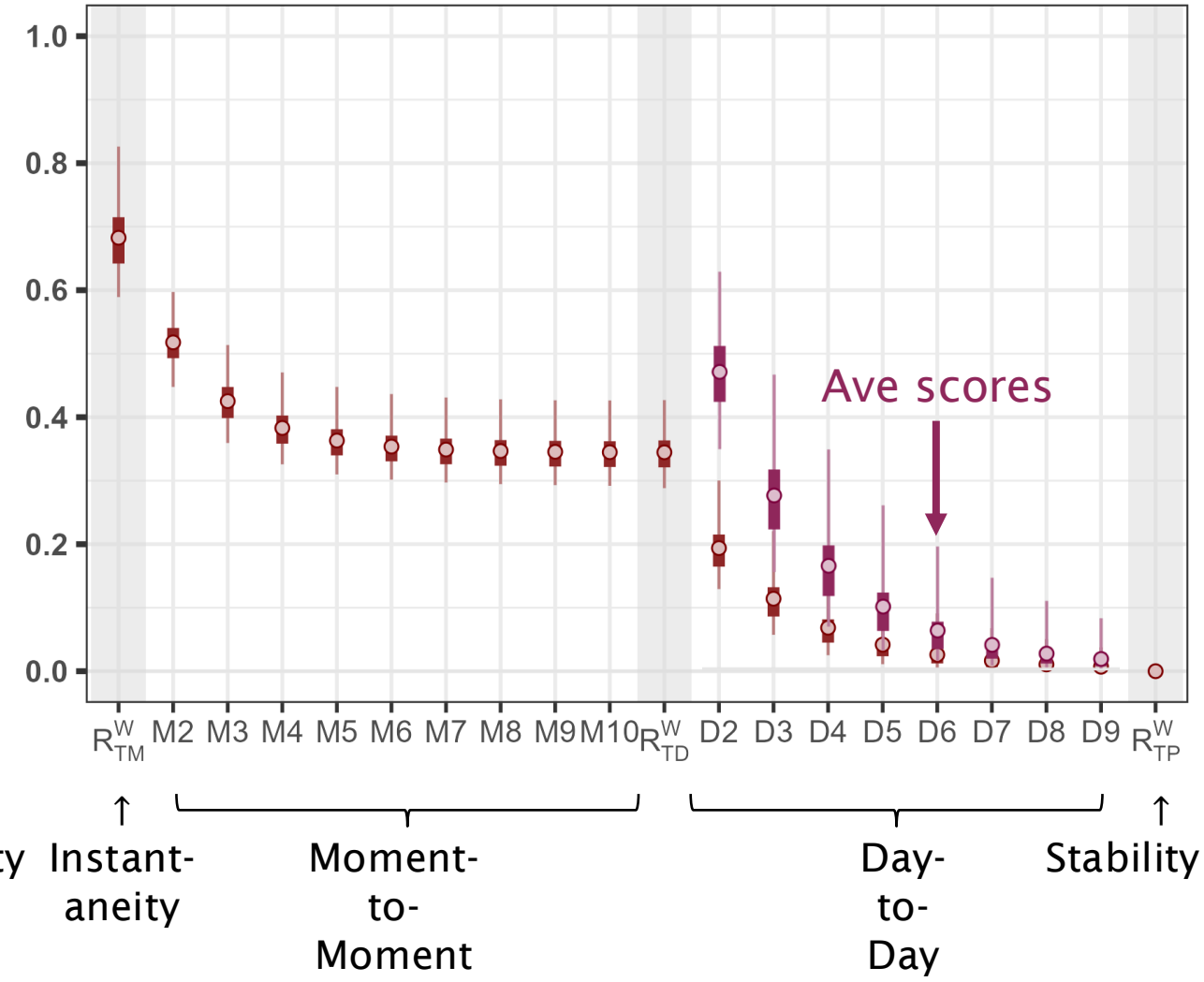


Estimated Reliability

Between-person



Within-person



Reliability for Different Models

- **M1:** Three-level model with two AR(1) processes

$$y_{ijk} = \beta + s_i + d_{ij} + \nu_{ij} + \omega_{ijk} + \epsilon_{ijk}$$

$$\begin{aligned}
s_i &\sim N(0, \sigma_s^2) \\
d_{ij} &\sim N(0, \sigma_d^2) \\
\nu_{ij} &\sim N\left(\phi_d \nu_{ij-1}, \frac{\tau_d^2}{1-\phi_d^2}\right) \\
\omega_{ijk} &\sim N\left(\phi_m \omega_{ijk-1}, \frac{\tau_m^2}{1-\phi_m^2}\right) \\
\epsilon_{ijk} &\sim N(0, \sigma_\epsilon^2)
\end{aligned}$$

- **M2:** Three-level model

$$y_{ijk} = \beta + s_i + d_{ij} + \epsilon_{ijk}$$

- **M3:** Two-level model with one AR(1) process

$$y_{it} = \beta + s_i + \omega_{it} + \epsilon_{it}$$

- **M4:** Two-level model (= **Classical test theory**)

$$y_{it} = \beta + s_i + \epsilon_{it}$$

Merge two time scales j & k into a single scale t

Between-person	Time span			
	Instantaneity	Moment-to-Moment	Day-to-Day	Stability
M1: Three-level LMM with two AR(1)	$\rho_0^B = R_{TM}^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_M^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 \phi_m^{\lvert k-k' \rvert}}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_D^B = \frac{\sigma_s^2 + \tau_d^2 \phi_d^{\lvert j-j' \rvert}}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_\infty^B = R_{TP}^B = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$
M2: Three-Level LMM	$\rho_\infty^B = \rho_M^B = R_{TM}^B = R_{TD}^B = \frac{\sigma_s^2 + \sigma_d^2}{\sigma_s^2 + \sigma_d^2 + \sigma_\epsilon^2}$		$\rho_D^B = \rho_\infty^B = R_{TP}^B = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_d^2 + \sigma_\epsilon^2}$	
M3: Two-Level LMM with one AR(1)	$\rho_0^B = R_{TM}^B = \frac{\sigma_s^2 + \tau_t^2}{\sigma_s^2 + \tau_t^2 + \sigma_\epsilon^2}$	$\rho_T^B = \frac{\sigma_s^2 + \tau_t^2 \phi^{\lvert t-t' \rvert}}{\sigma_s^2 + \tau_t^2 + \sigma_\epsilon^2}$		$\rho_\infty^B = R_{TP}^B = \frac{\sigma_s^2}{\sigma_s^2 + \tau_t^2 + \sigma_\epsilon^2}$
M4: Two-Level LMM (= CTT)	$\rho = R_T = \frac{\sigma_s^2}{\sigma_s^2 + \sigma_\epsilon^2}$			

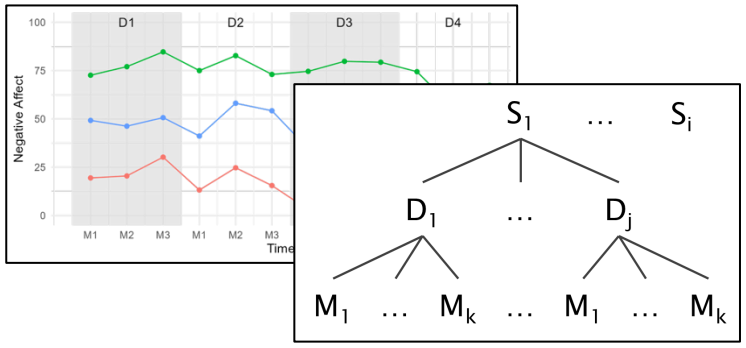
$R_{TD}^B = \frac{\sigma_s^2 + \sigma_d^2 + \tau_d^2}{\sigma_s^2 + \sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$

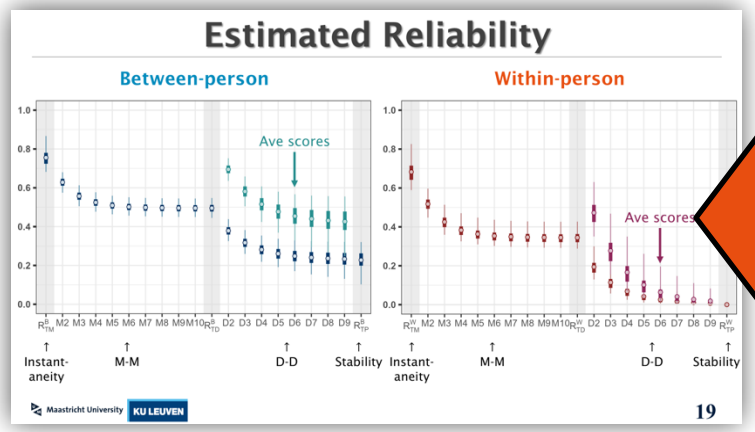
Within-person	Time span			
	Instantaneity	Moment-to-Moment	Day-to-Day	Stability
M1: Three-level LMM with two AR(1)	$\rho_0^W = R_{TM}^W = \frac{\sigma_d^2 + \tau_d^2 + \tau_m^2}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_M^W = \frac{\sigma_d^2 + \tau_d^2 + \tau_m^2 \phi_m^{\lvert k-k' \rvert}}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	$\rho_D^W = \frac{\tau_d^2 \phi_d^{\lvert j-j' \rvert}}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$	-
M2: Three-Level LMM	$\rho_M^W = R_{TM}^W = R_{TD}^W = \frac{\sigma_d^2}{\sigma_d^2 + \sigma_\epsilon^2}$		-	
M3: Two-Level LMM with one AR(1)	$\rho_0^W = R_{TM}^W = \frac{\tau_t^2}{\tau_t^2 + \sigma_\epsilon^2}$	$\rho_T^W = \frac{\tau_t^2 \phi^{\lvert t-t' \rvert}}{\tau_t^2 + \sigma_\epsilon^2}$		-
M4: Two-Level LMM (= CTT)	-			

$$R_{TD}^W = \frac{\sigma_d^2 + \tau_d^2}{\sigma_d^2 + \tau_d^2 + \tau_m^2 + \sigma_\epsilon^2}$$

Discussion

Data





Model

Account for
the **serial correlation**
and
the **multilevel structure**
(S > D > M)

L3: Sub-level L2: Day-level L1: Moment-level

$$y_{ijk} = \beta + s_i + d_{ij} + \nu_{ij} + \omega_{ijk} + \epsilon_{ijk}$$

Day-level AR(1) process

$$\nu_{ij} = \phi_d \nu_{ij-1} + \eta_{ij}$$
$$\eta_{ij} \sim N(0, \tau_d^2/(1-\phi_d^2))$$

Moment-level AR(1) process

$$\omega_{ijk} = \phi_d \omega_{ijk-1} + \eta_{ijk}$$
$$\eta_{ijk} \sim N(0, \tau_m^2/(1-\phi_m^2))$$

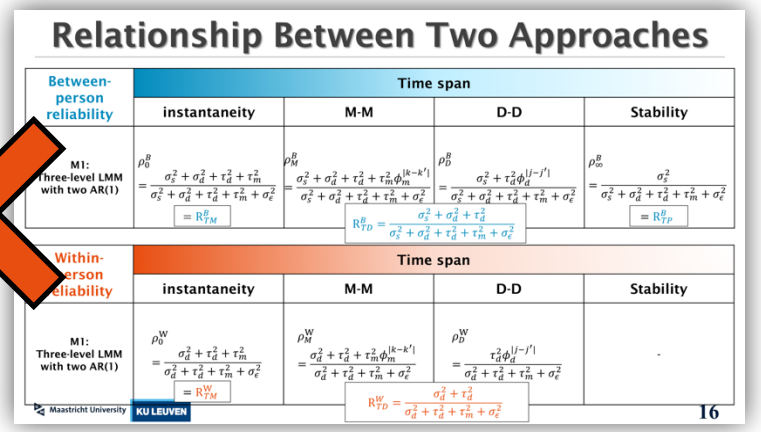
Measurement Error

$$\epsilon_{ij} \sim N(0, \sigma_\epsilon^2)$$

Reliability

Correlation Approach

Variance Partitioning Approach



Discussions and Limitations

- Reliability coefficients are largely dependent on the **model assumptions**
  - Different types of serial correlation processes
  - Potential covariates
  - Heterogeneity of variance:
    - Personal-specific parameters ⇒ personal reliability
    - Time-specific parameters ⇒ time-varying reliability
- Reliability for **a single measurement/item with a continuous scale**.
  - Other approaches.
    - E.g., Schuurman & Hamaker (2019); Dejonckheere et al. (2022)
  - Multiple-item reliability?
    - E.g., Castro-Alvarez et al., (2025); Hunter (2026)
  - Bounded data for the measurement (0-100), binary data?
    - E.g., Molenberghs et al., (2007); Vanbelle et al.,(2025)

Take-home message

Reliability is not a fixed property of an instrument but changes with time and the hierarchical level.

Accepted manuscript

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